

HARNESSING THE SOLAR-POWERED CAR AND SOLAR CHARGE CONTROLLERS

WRITTEN TO FULFILL THE REQUIREMENT FROM THE SCHOOL



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A. Introduction

I. Research Background

The highest recorded average temperature on Earth occurred on July 4, 2023, with as high as 17.18 °C (Paddison, 2023). This occurrence is the major effect of greenhouse gases, which work by trapping the sun's heat and preventing it from escaping into space, thereby increasing the Earth's temperature. Taken from the EPA (United States Environmental Protection Agency), an average passenger vehicle releases approximately 4.6 metric tons of carbon dioxide annually ("Greenhouse Gas Emissions from a Typical Passenger Vehicle," n.d.). Just imagine how much if combined by every passenger. Moreover, the combustion of gasoline and diesel fuel produces harmful byproducts such as nitrogen dioxide, carbon monoxide, and benzene ("Reducing Air Pollution From Cars," n.d.). The increase in greenhouse gases and global warming is acknowledged, and this research focuses on the utilization of solar energy, especially in the vehicle industry, along with the development of solar charge controllers that regulate the flow of current and voltage to the battery preventing damaging it.

The background of this research revolves around the use of renewable energies, particularly solar energy. In H1 2023, the use of solar panels had increased by 153% in China, followed by Germany with 102% (Feldman, 2023). While the use of solar energy is widespread today, its application in vehicles remains minimal, possibly influenced by various factors. This project investigates whether solar panels alone can effectively power a vehicle. My team decided to manufacture a model of solar-powered car.

II. Research Focus

The key objective is to include the use of one of the most accessible renewable energies in the vehicle industry, which is solar energy. Not only that, the application of solar charge controllers is also in investigation.

III. Research Questions

1. How can we make sure that the battery is not overcharged by the solar panels?
2. How can we arrange solar panels, rechargeable batteries, and dynamos in a manner that optimizes the performance of the car?

IV. Research Objectives

1. Designing and applying the solar charge controller in the circuit.
2. Set up the circuit and determine the role of each component.

V. Research Benefits

This study aims to raise awareness about the global warming phenomenon and encourages individuals to transition towards renewable energies, specifically solar energy. It provides information about the implementation of solar energy in the vehicle industry for further application. In addition, the use of solar charge controllers will broaden the knowledge of this

specific electrical component and its purpose. Alongside, serving as an educational tool in increasing knowledge in physics subjects about electrical components in use and some complex circuit designs inside the solar charge controller. Thereby giving exposure to minor electrical components, such as operational amplifiers, resistors, inductors, capacitors, et cetera.

VI. Research Coverage

This research will be conducted in the field with the following limitations:

- (1). The manufacture of the car is only up until the model, not until the real-life size or prototype.
- (2). The design of the solar charge controller, including the buck converters and other components necessary, is limited up until the drawing and theoretical calculations. Practical experiments were also restricted to virtual simulations conducted using a website called [Multisim Live Online Circuit Simulator](#), owing to the absence of certain electrical components.

B. Literature Review

I. Background Theory

I.1. Solar Energy and Solar Panels

Solar energy stands out as one of the most accessible forms of renewable energy. It refers to the radiation from the Sun, capable of producing heat, triggering chemical reactions, or generating electricity (Ashok, 2024). The only way to utilize solar energy is by converting it into electricity, a process known as the photovoltaic effect.

The photovoltaic effect involves converting light energy into electricity, generating voltage and current, following the photoelectric effect principle. The mechanism is that the excited charge carriers are excited and remain within the material, allowing the generation of voltage and current. Excited electrons and holes create electron-hole pairs, leading to the formation of the P-N (Positive-Negative) junction between the semiconductor materials. This junction contributes in the separation of electron-hole pairs, with the electric field causing electrons to move toward the N-junction and holes toward the P-junction. This movement results in the generation of voltage and electric current.

As previously mentioned, solar energy can be captured using solar panels. For them to be effective, the material used must have a low work function characteristic. This allows electrons to be excited with less energy from the photons. Another factor impacting panel efficiency is the frequency and intensity of light, with frequency determining the energy carried per photon and intensity influencing the rate of incoming photons. The Earth's average intensity when directly facing the Sun is about 1360 watts per square meter (Przyborski, n.d.).

I.2. Solar Charge Controller

Solar charge controller serves the purpose of preventing battery overcharging by managing the voltage and current flow from the solar panel to the battery (Grob, 1989). The type used is MPPT (Maximum Power Point Tracking). MPPT controllers utilize advanced technology to enable the solar array to function at its peak power point, ensuring optimal current and voltage. This results in the conversion of surplus energy into additional power for your batteries. These controllers accomplish this by converting excess voltage into increased current, directing more energy into the batteries. Despite their higher initial cost, MPPT controllers offer long-term savings by maximizing the energy harvested from the solar panels. They prove particularly beneficial when battery levels are low, swiftly recharging the battery by converting surplus voltage into additional current.

I.3. Operational Amplifiers (Op-amp)

The primary role of an operational amplifier (op-amp) is to significantly magnify the difference between its two input terminals and produce an output based on this difference. If the positive input (V^+) voltage is higher than the negative input (V^-), the op-amp will amplify and generate a positive output. Conversely, if the voltage at (V^-) is higher, the op-amp will produce an amplified negative output.

The op-amp has several functions, in this case, we are using op-amp as a comparator in a circuit. The basic comparator circuit is to connect a couple of resistors to the inverting input and another to the non-inverting input. Using the concept of the potential divider, the circuit will then be designed to have either non-inverting or inverting input higher in voltage to amplify the positive or negative output. In this project, the op-amp used is LM358, which consists of two separate, internally frequency-compensated operational amplifiers with high gain. According to Diodes Incorporated, the company that manufactured them, these amplifiers are specifically engineered to function effectively using a single power supply across a broad range of voltage values ("LOW POWER DUAL OPERATIONAL AMPLIFIERS", n.d.).

I.4. Inductors

Inductors work by using the principle of electromagnetic induction. When current flows through the coil, it will create a magnetic field that will store energy. If the current flowing through is constant, then there will be current stored in the form of a magnetic energy starting from low to constant value. When the current is cut off, the magnetic field is reduced gradually and the magnetic energy is converted back to electrical energy. The keyword here is gradually, so the voltage-time graph generated would be smoothed in the case of alternating voltage or current.

I.5. Capacitors

A Capacitor is a component that stores capacitance, which refers to the amount of charge stored on the plates of a capacitor per unit of potential difference across those plates. A capacitor consists of two parallel plates, which store an equal amount of electrons and protons. When a capacitor is connected to a power source, the electrons will be attracted by the positive polarity

and move to the opposite plate, making one plate more negative than the other, this produces a potential difference between the plates. This process is called charging because now there is a potential difference and capacitance formed and ready to be calculated. In contrast, when the power source is removed, the electrons from one plate will go back to the opposite plate because the protons attract them, this will happen until both plates have zero potential difference, this action is called discharging. Furthermore, capacitors can store charge and energy at the same time this is due to the action of stripping away electrons from one plate from the nucleus, the cells should do work and push that electron toward another plate. The purpose of a capacitor is also to smooth out voltage because the action of charging an discharging takes time and not an instant action.

II. Relevant Research

The following are relevant research with similar variables that are tested in this research paper:

1. This paper by T. Koyuncu talks about how the operational efficiency of a semi-flexible monocrystalline silicon solar panel utilized in the "Firat Force" solar-powered car and the "Commagene" solar-powered minibus. The underlying reason for this research was because of the increase in the amount of greenhouse gases available. Therefore, photovoltaic (PV) panels are under investigation, especially in the vehicle industry. They also measure the required PV area in order to power up the DC motor power, to estimate the power required for any vehicle, factors such as the average incident solar radiation and daytime conditions in the city (averaged over approximately 40 years), vehicle speed, efficiency of the PV panels, road characteristics and slope, passenger capacity, necessary daily travel distance, and the efficiency of all equipment were taken into account.

III. Research Framework

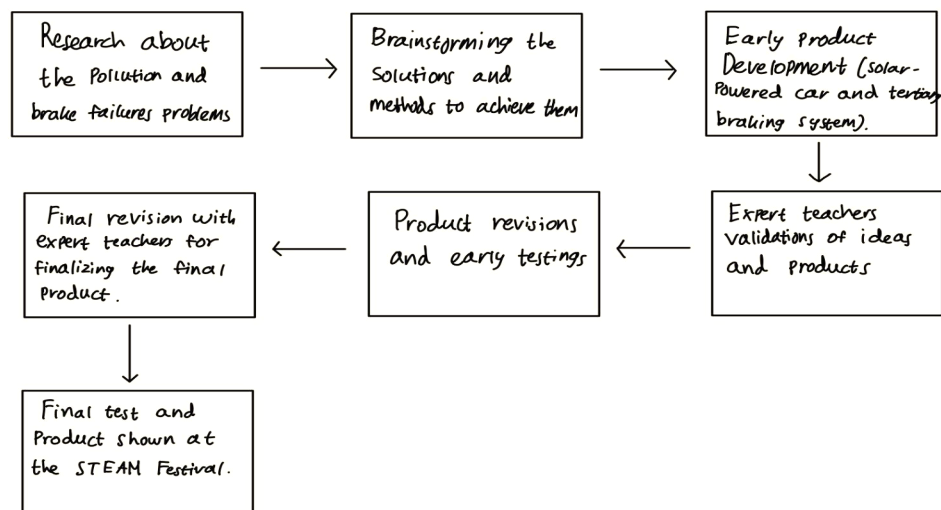


Figure 1. Research Flowchart

C. Methodology

I. Research Goals

This research aims to acknowledge the function of solar energy in the vehicle industry along with the implementation and development of solar charge controllers.

II. Time and Location

This research will be conducted in SPK SMAK PENABUR KELAPA GADING during and after school hours.

III. Data Resources

Data was collected mostly from the internet or online journals. Most theories and technical terms were taken from the Cambridge Physics Coursebook and Basic Electronics by Bernard Grob. Numerical data and calculations were taken from hands-on experiments in the Physics laboratory or virtual experiments.

IV. Research Methods and Procedures

1. Car

The mechanism involves the solar panels directly powering the two dynamo motors in the rear, known as Rear-Wheel Drive (RWD). Additionally, the car is designed to operate with or without sunlight, combining rechargeable batteries to store energy. In essence, the entire circuit will include the solar panel connected in parallel to the dynamo motors, solar charge controller, and rechargeable batteries. Parallel connection is crucial because the voltage stored needs to be the same for the whole circuit, ensuring the solar panels are strong enough to power the entire circuit. In addition, the solar charge controller will be connected between the batteries and the solar panels to serve its purpose.

2. Solar Charge Controller

2. A. Solar Charge Controller

The final circuit will include a solar charge controller, which will act as a stopper for the current to flow when the battery is already full. The type used is MPPT (Maximum Power Point Tracker). The actual diagram of MPPT is very complicated. However, the main part of MPPT being used is the buck converter, which will be explained more in detail. It functions to chop down the voltage by using an automatic switch and smoothed to become direct voltage. In this way, the excessive voltage coming from the solar panel can be stepped down depending on the maximum voltage intake of the battery.

2. B. Step-Down Voltage Converter

The first component of the buck converter is the step-down voltage converter (Figure 3). There are multiple components in use, such as a switch, diode, inductor, capacitor, and resistor. Components here each have different roles, starting with the switch, which is used to step down the frequency of the voltage coming in. If the switch is open and closed at a certain

frequency, the voltage-time graph would be chopped with a maximum value of voltage when the switch is closed and zero when open. The higher the frequency of the switch, the lower the duty cycle will be, so the lower voltage will be produced. The addition of inductors and capacitors is for smoothing the stepped-down voltage to the battery to prevent damaging it. As a final input, the diode works to create the correct path for electrons and prevent any electrons from accumulating only in one place and causing failure.

Furthermore, the purpose of the circuit above is only to step down the input voltage, there has not been a way where there can be a sensor that stops the current when the battery is fully charged. This is where the full design of the buck converter comes in (check Figure 4). Numbers are not yet provided due to the need for calculations almost in every step of this circuit to reach the desired outcome.

The mechanism starts from the voltage exerted by the solar panel following the photovoltaic effect principle. Then it is regulated through three different voltage regulators (LM1117), with V_1 being the highest and reduced in V_2 and V_3 . Capacitors are connected parallel next to every voltage regulator to smooth out the voltage coming out. The V_1 is then connected to become the supply voltage of the second op-amp and the gate of the P-Mosfet. The V_2 is connected to the supply voltage of the first amplifier, which acts as the differential amplifier, and the triangle wave generator, which later connects to two other op-amps as supplying voltage. The last voltage, V_3 , acts as the reference voltage to become the input voltage in the non-inverting input in the differential amplifier.

2. C. Differential Amplifier

The mechanism then flows onto the differential amplifier, it uses the principle of closed-loop gain. As we know the voltage is available in both inputs, so it is a combination of inverting and non-inverting op-amp. Therefore, the output voltage is given by the formula of $A(V^+ - V^-)$, where A is the gain. In open-loop gain, the gain is determined by the specification of the type of op-amp used, or usually infinity. However, in this case, the gain can be determined by $\frac{R_f}{R_i}$, where R_f is the feedback resistor, and R_i is the input resistor, the derivation will be provided in the findings. In order for the output to always be positive DC, the inverting input always has to be smaller than the non-inverting. In the opposite case, the output will immediately become grounding because the negative supplying voltage is set to ground.

In the case of this circuit, the non-inverting input is always constant, coming from V_3 . The inverting input, however, is always changing depending on the voltage exerted by the rechargeable battery. The battery is connected to a potential divider, where there is a resistor and an automatic potentiometer that increases its resistance whenever the battery is fully charged, and the other way around.

2. D. Triangle Wave Generator

The output voltage from the differential amplifier is then connected to the inverting input of the next op-amp. It is compared to a triangle wave generator, which is connected to the non-inverting input. The mechanism of the triangle wave generator includes similar components as the rest of the circuit, including op-amps. There are three steps, starting from

an astable multivibrator, as the square wave generators, then it is fed to an integrator, which will generate it to become triangle waves.

Starting from the astable multivibrator (Figure 5). The mechanism starts at both the inverting and non-inverting input grounded. However, the output is fed back to both inputs, the non-inverting input is connected to a potential divider, and the inverting input is connected to an adjustable potentiometer and a capacitor. Assuming that the output voltage is positive at the beginning, the electrons from the capacitor will be attracted towards the output voltage, this means the current is flowing toward the capacitor, charging it. As it is charging, the voltage at the inverting node of the op-amp starts to build up and become higher than the non-inverting input (assuming it is an ideal op-amp, so no current flowing to the inputs). The output will now be negative, so the capacitor will start discharging toward the negative voltage, the voltage at the inverting node will now decrease and the voltage at the non-inverting node will start to build up (assuming the R_2 must be bigger than R_1 to generate higher voltage). In this way, the output voltage will become positive again.

Furthermore, the next step is the integrator, it will act as the converter from square-wave voltage to triangle wave voltage. The concept used in this component is integration, it acts to measure the area under the curve as the output. However, the difference is that the output is inverted due to the derivation of the formula. Therefore, if the input is a positive voltage, the output will start decreasing gradually, but when the input is changed to negative, the output will start increasing gradually, as in the triangle wave graph (Figure 6). The working principle of the integrator is similar to the differential amplifier, the inverting op-amp is trying to keep both of its inputs the same. However, the difference is only that one of the resistors is replaced by a capacitor.

2. E. PWM (Pulse Width Modulation) Generator and Transistor Diode

In the next step, the DC output voltage from the differential amplifier and the triangle wave voltage from the triangle wave generator are fed to another op-amp, in which the non-inverting input is from the differential amplifier and the inverting input is from the triangle wave generator. This op-amp acts as a comparator, which compares both inputs and discharges the supplying voltage, depending on the higher input, as the output. The output will form PWM because the triangle wave voltage will always be changing and become higher and lower than the DC voltage. This results in the output always changing, which are the positive supply voltage and the grounding. More clearly can be seen in the voltage-time graph (Figure 7). Furthermore, the duty cycle of the PWM output can be changed depending on the value of the DC voltage from the differential amplifier. If the value increases, the duty cycle will decrease, because the duration of the higher value of the triangle wave voltage will be lower, so the output will be negative supply voltage (grounding) for a longer duration. In contrast, if the value of the DC voltage decreases, the duration of the higher value of the triangle wave voltage will be longer, hence increasing the duration of the positive supply as the output.

Lastly, the transistor diode acts as a switch. It connects the supply voltage, which is V_s , the PWM voltage, and the step-down voltage converter. This component serves to match the duty cycle and frequency from the PWM to the supply voltage and relay it to the step-down voltage converter. For instance, if the supply voltage is $12V$ with the PWM duty cycle of 50%

and frequency of $1kHz$, then there will be a chopped voltage with a maximum of $12V$ and a lowest of $0V$ with $1kHz$ of frequency. However, this is assuming that all components are ideal, in real cases, there might be input offset voltage, affecting the values to not be precise by a few millivolts.

V. Diagrams and Graphs

V. 1. Whole Circuit Design

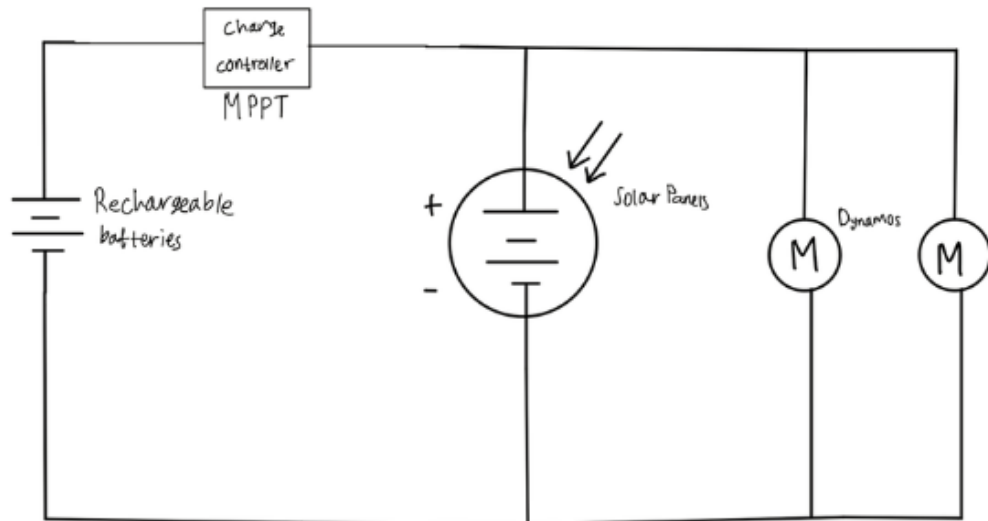


Figure 2. The Circuit Design For The Energy System of The Car

V. 2. DC Step-Down Voltage Converter

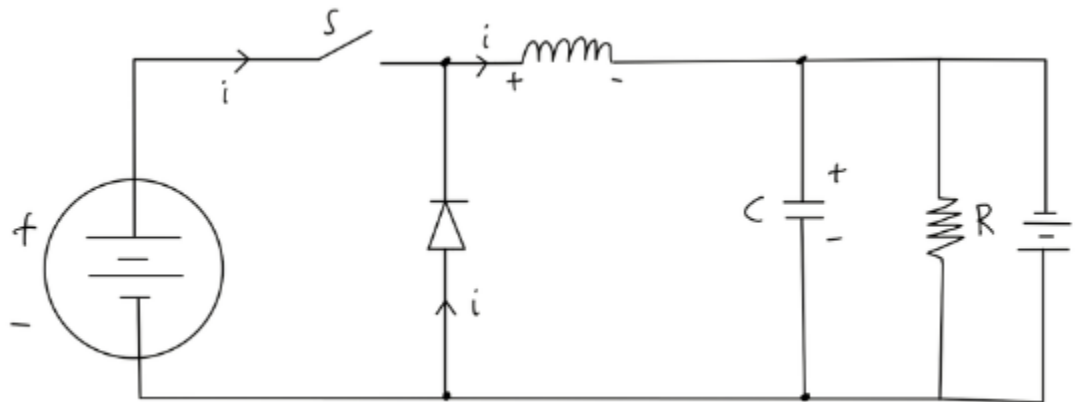


Figure 3. The Design Diagram of The Step-Down Voltage Converter

V. 3. Buck Converter

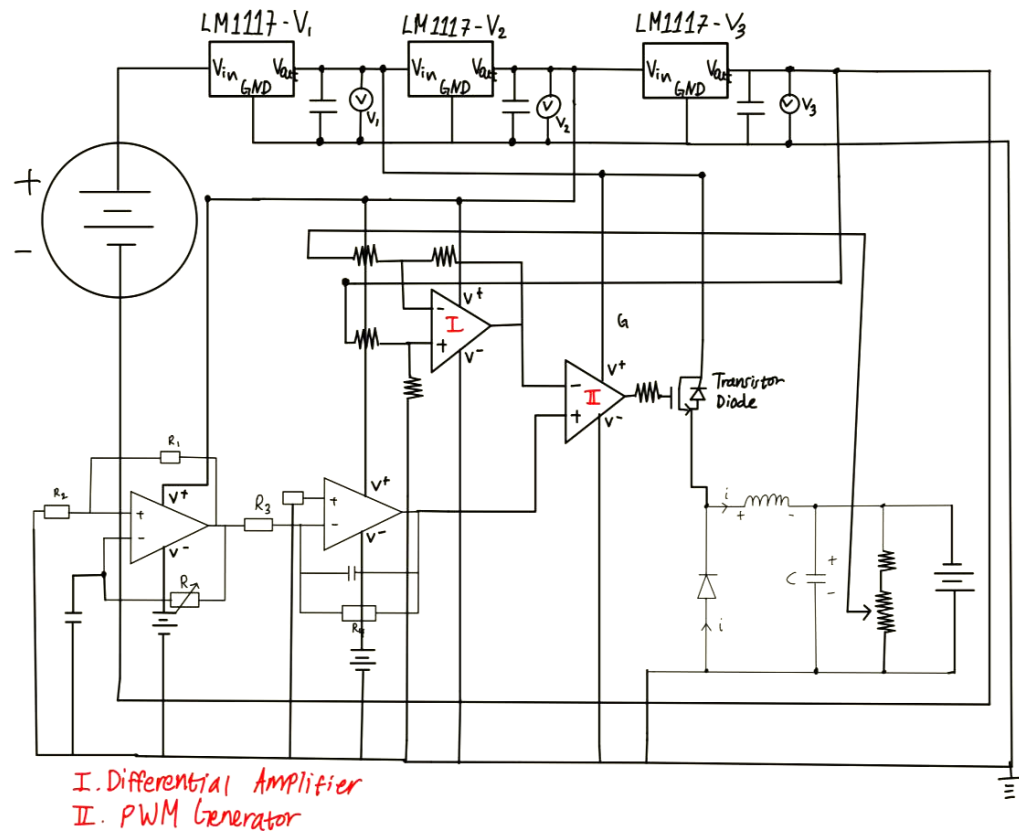


Figure 4. The Complete Design Diagram of Buck Converter

V. 4. Triangle Wave Generator

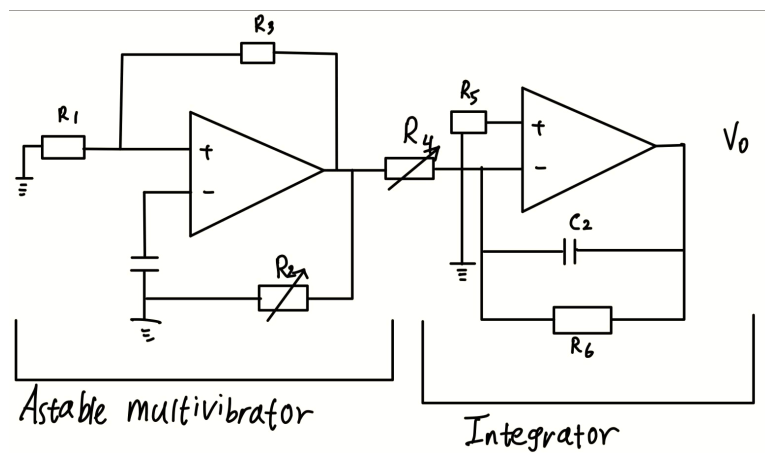


Figure 5. Triangle Wave Generator

V. 5. Triangle Wave Graph

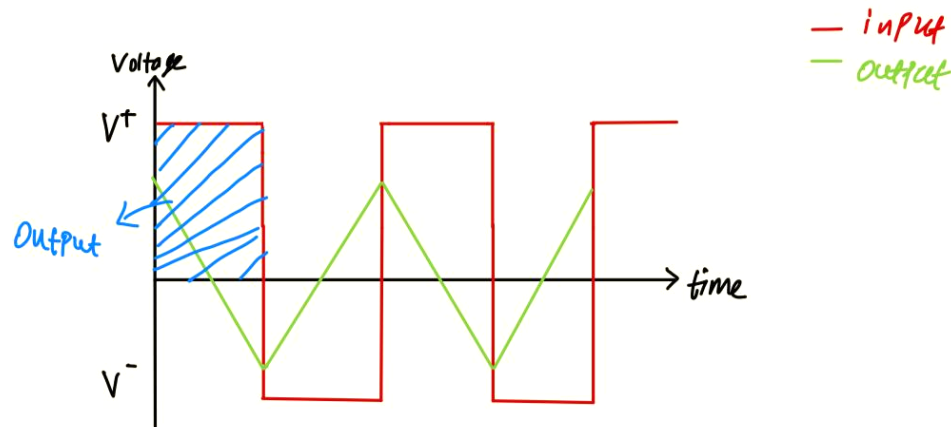


Figure 6. Triangle Wave Graph

V. 6. PWM Voltage Input-Output Graph

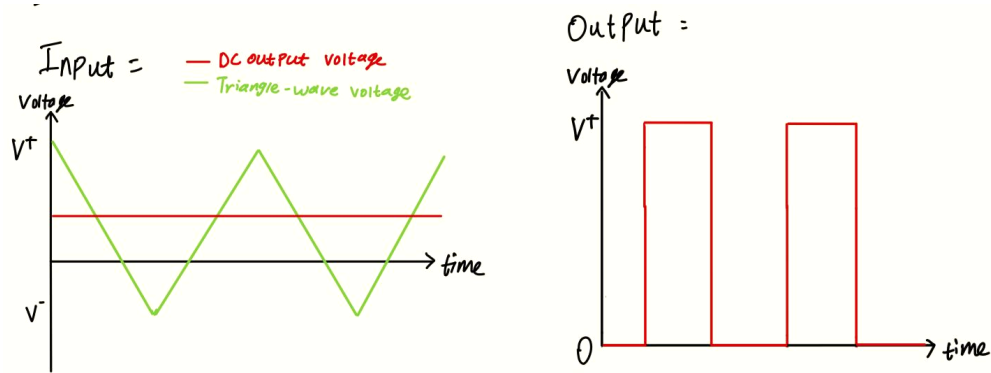


Figure 7. PWM Voltage Input-Output Graph

D. Findings

I. Theoretical Calculations and Virtual Experiments

In this part, the calculations and virtual experiments of components in the buck converter will be demonstrated. All the calculations are in the assumption of using all ideal components, and the experiment results are near to real life.

1. Triangle Wave Generator

The first component being conducted for the experiment is the triangle wave generator. However, before stepping into the lab, calculations are required to reach the desired outcome. The objectives are the frequency, output voltage from the astable multivibrator, value of resistors and capacitance, and the final output voltage. Assuming that the desired outcome of the triangle wave is $\pm 3.75\text{V}$ with a frequency of 1kHz. As a

note, the adjustable components in this circuit are only the potentiometers and supply voltage.

Calculation:

Astable Multivibrator

- Period (T) = $2RC \ln\left(\frac{1+\beta}{1-\beta}\right)$

$$\beta = \frac{R_2}{R_2+R_1}$$
 If $f = 1 \text{ kHz}$, then $T = 1 \text{ ms}$
- Assumptions:
 $C = 0.1 \mu F$
 $R_1 = R_2 = 10 \text{ k}\Omega$
 $V_{cc} = 15V$
 $V_{ee} = -15V$
- Finding R_2 of potentiometer:

$$1 \times 10^{-6} = 2R(0.1 \times 10^{-6}) \ln\left(\frac{1+0.5}{1-0.5}\right)$$

$$R_2 = 4.55 \text{ k}\Omega$$

Integrator

- The period of the integrator should match with the astable multivibrator.
 $T = R_4 C_2 = 1 \text{ ms}$
- Assumption:
 $C_2 = 0.01 \mu F$
- Finding R_4 :

$$R_4 = \frac{1 \times 10^{-3}}{1 \times 10^{-6}} = 100 \text{ k}\Omega$$
- Finding the amplitude of the triangle wave voltage:

$$V_{out} = - \frac{1}{RC} \int_0^{\frac{T}{4}} V_{in} dt, \text{ where } R \text{ is } R_4 \text{ and } C \text{ is } C_2$$

Derivation of the formula:

Current through capacitor =

$$Q = CV$$

$$V = \frac{Q}{C}$$

$$V = \frac{1}{C} \int I dt$$

$$\frac{dV}{dt} = \frac{1}{C} I$$

$$I = C \frac{dV}{dt}, \text{ where } \int I dt : \text{charge accumulated on the capacitor}$$

Current through R_3 = Current through capacitor

$$\frac{V_{in}}{R} = C \frac{d(0-V_o)}{dt}$$

$$\int \frac{V_{in}}{R} dt = C \int \frac{d(-V_o)}{dt} dt$$

$$\frac{1}{RC} \int V_{in} dt = - \frac{dV_o}{dt}$$

$$V_o = - \frac{1}{RC} \int_b^a V_{in} dt \text{ (Proven)}$$

$a = \frac{T}{4}$, $b = 0$ because the time it takes for the voltage to reach peak from zero is a quarter period.

Finding V_o , when V_{in} is $15V$:

$$V_o = - \frac{1}{100 \times 10^3 \times 0.01 \times 10^{-6}} \int_0^{\frac{1 \times 10^{-3}}{4}} 15 dt$$

$$V_o = - 1000 \times \left[\left(15 \times \frac{1 \times 10^{-3}}{4} \right) - (15 \times 0) \right]$$

$$V_o = - 3.75V$$

Finding V_o , when V_{in} is $-15V$:

$$V_o = - \frac{1}{100 \times 10^3 \times 0.01 \times 10^{-6}} \int_0^{\frac{1 \times 10^{-3}}{4}} 15 dt$$

$$V_o = - 1000 \times \left[\left(- 15 \times \frac{1 \times 10^{-3}}{4} \right) - (15 \times 0) \right]$$

$$V_o = 3.75V$$

Virtual experiment:

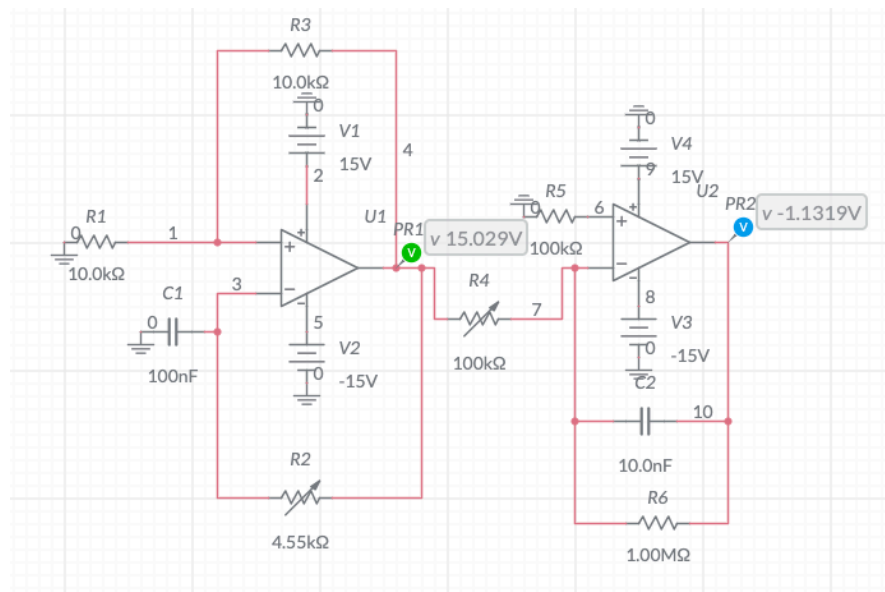


Figure 8. Triangle Wave Generator Virtual Experiment

Result:

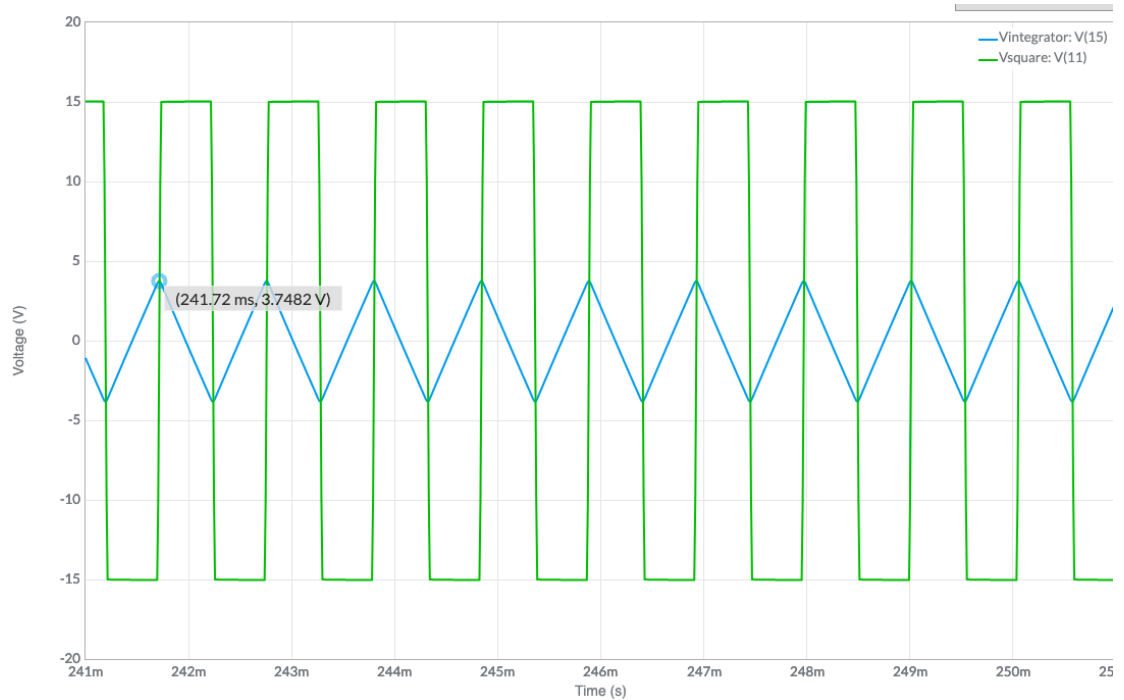


Figure 9. Triangle Wave Generator Result

Amplitude: $3.748V$ and $-3.748V$

Period (T): $1.0 \pm 0.1 \text{ ms}$

Conclusion:

The potentiometer R and R_3 is inversely proportional to the frequency of the wave, hence the frequency is directly proportional to the amplitude of the triangle wave voltage or output of the integrator. In the experiment, the numbers are not true to the calculation due to the non-ideal components.

2. Differential Amplifier

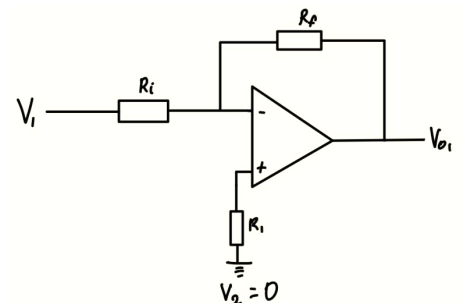
The next component being conducted is the differential amplifier. There needs to be calculations on how this component can determine that the load battery is fully charged or not. This is due to the voltage from the load fed back to the inverting input.

Calculations:

- Voltage output formula: $A(V^+ - V^-)$, where $A: \frac{R_f}{R_i}$
- Derivation of formula:
 1. If non-inverting input is grounded:

$$\text{Gain} = -\frac{R_f}{R_i} = \frac{V_{out1}}{V_1}$$

$$V_{out2} = -\frac{R_f}{R_i}V_1$$



2. If inverting input is grounded:

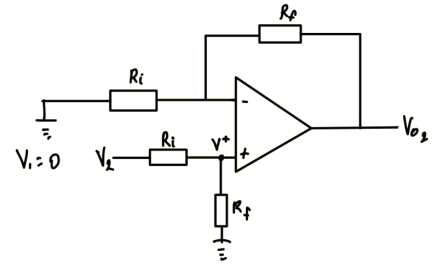
$$\text{Gain: } 1 + \frac{R_f}{R_i}$$

$$V_{out_2} = \left(1 + \frac{R_f}{R_i}\right) V^+$$

$$V^+ = \frac{R_f}{R_i} V_2$$

$$V_{out_2} = \left(1 + \frac{R_f}{R_i}\right) \times \frac{R_f}{R_i + R_f} V_2$$

$$V_{out_2} = \left(\frac{R_f}{R_i}\right) V_2$$



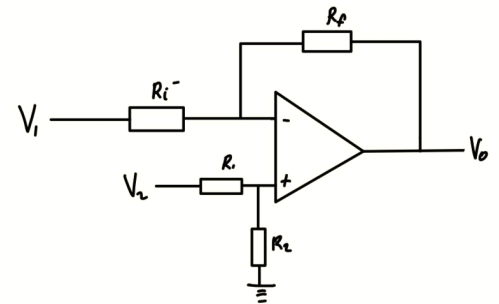
3. Both inputs have voltage:

$$V_{out} = V_{out_1} + V_{out_2} = A(V_2 - V_1)$$

$$V_{out} = \left(-\frac{R_f}{R_i}\right) V_1 + \left(\frac{R_f}{R_i}\right) V_2$$

$$V_{out} = \frac{R_f}{R_i} (V_2 - V_1) = A(V_2 - V_1)$$

$$A = \frac{R_f}{R_i}$$



- Voltage output:

$$V_{out} = \frac{R_f}{R_i} (V_2 - V_1)$$

$$V_{out} = \frac{1}{2} (2.5 - 2.3)$$

$$V_{out} = 100mV$$

To ensure the output voltage doesn't exceed the desired, the V_{cc} of the op-amp is also set to $100mV$.

Result:

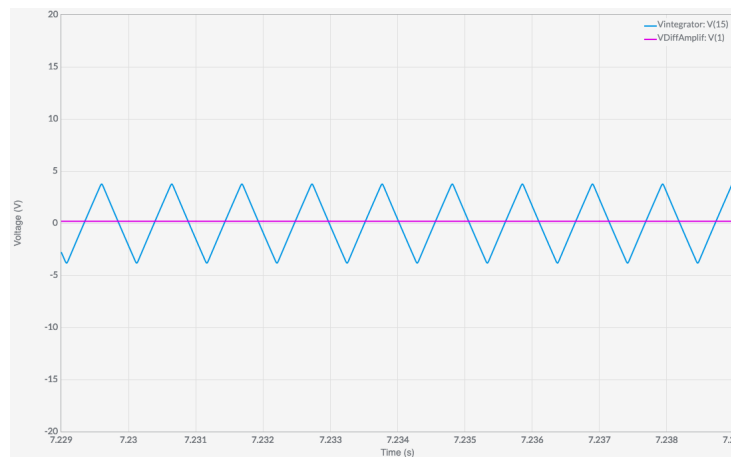


Figure 10. Differential Amplifier Output Result

From the data above, we can calculate the duty cycle of the PWM Generator. This is done by subtracting the two durations of the value of the triangle wave, which are the ones above and below the DC Voltage.

$$\text{Duty cycle } (D) = \frac{\text{Time ON}}{\text{Period}} \times 100\%$$

$$\text{Time ON/Above: } 7.2350 - 7.23455 = 0.45\text{ms}$$

$$\text{Time OFF/Below: } 7.23555 - 7.2350 = 0.55\text{ms}$$

$$D = \frac{45}{100} \times 100 = 45\%$$

3. Full Circuit

The full circuit includes the triangle wave voltage in the non-inverting input and the DC voltage from the differential amplifier to the inverting input of the PWM generator. The PWM signal is then fed to the transistor diode, in which the input voltage is from the voltage regulator. The output of the transistor diode is then sent to the step-down DC voltage to produce a smoothed stepped-down DC voltage. Furthermore, the average current from the battery is stepped up, the experiment will validate it.

Calculations:

- Duty cycle (D) = 45%
- Input voltage: 12V
- Output voltage: $12 \times 0.45 = 5.40\text{V}$

Virtual experiment:

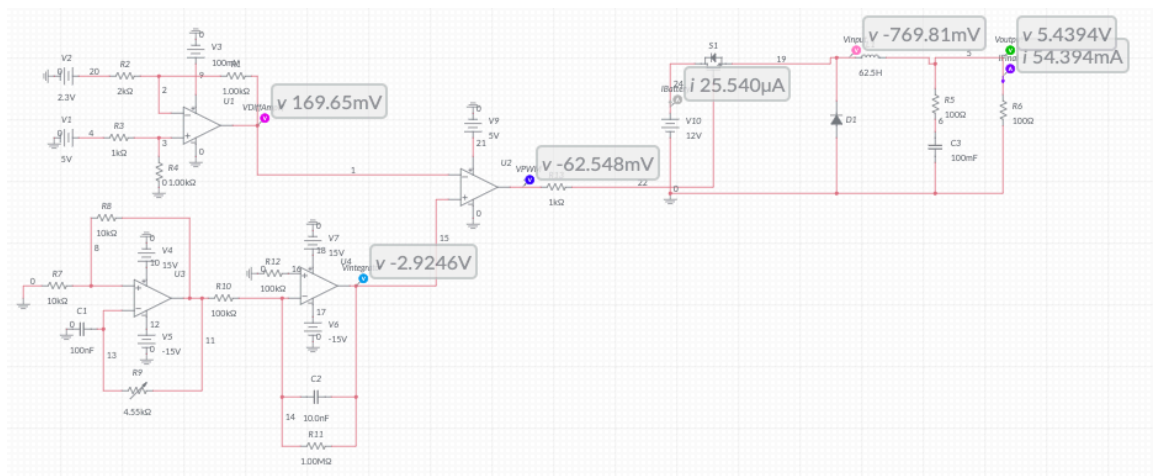


Figure 11. Full Circuit Design

Results:

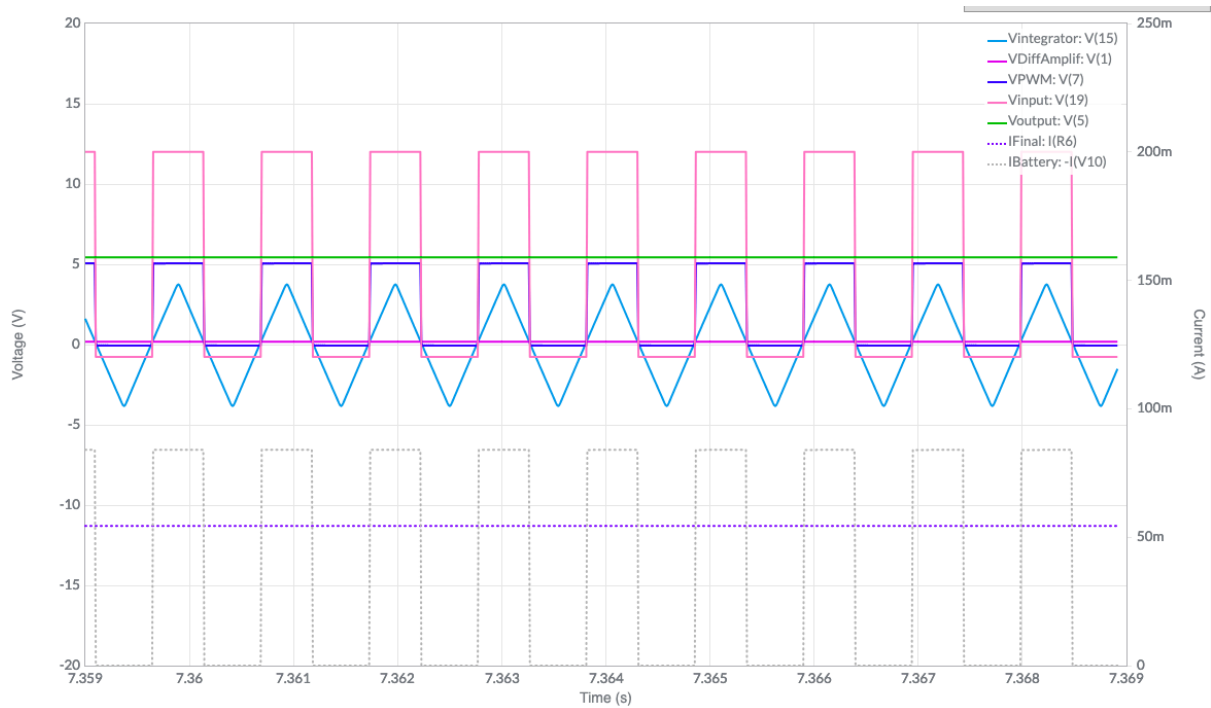


Figure 12. Complete Result of Voltage and Current From Final Circuit

Output voltage: $5.439V$

Output current: $\frac{5.439}{100} = 54.39mA$

Link to full circuit:

<https://www.multisim.com/content/cagRwxRpNRwktUzCSyPbm/buck-converter/open/>

E. Discussions and Conclusion

Discussions:

After a few experiments using different values of the components, it was found that increasing the inductance of the inductor will smooth the output voltage more. Then, increasing the capacitance and the current going through the capacitor will decrease the time it takes for the output voltage to reach the stable value. Furthermore, In real situations, the voltage exerted out from an op-amp might not respect the V_{cc} and V_{ee} already set up. For instance, the differential amplifier gives out $169.65mV$ when the V_{cc} is only $100mV$. Then, the PWM generator gives a value of $-62.55mV$ when the V_{ee} is set grounded. This is usually called input offset voltage.

Conclusion:

In conclusion, a solar charge controller is essential in any electrical circuit involving a solar panel and a rechargeable battery. Its function is to regulate excessive voltage to the battery and prevent damage. Although the circuit design and diagram function correctly, further investigation and experimentation are needed to prevent offset voltage from emerging. Additionally, a method must be devised for the rechargeable battery to provide feedback to the differential amplifier, indicating when it is fully charged.

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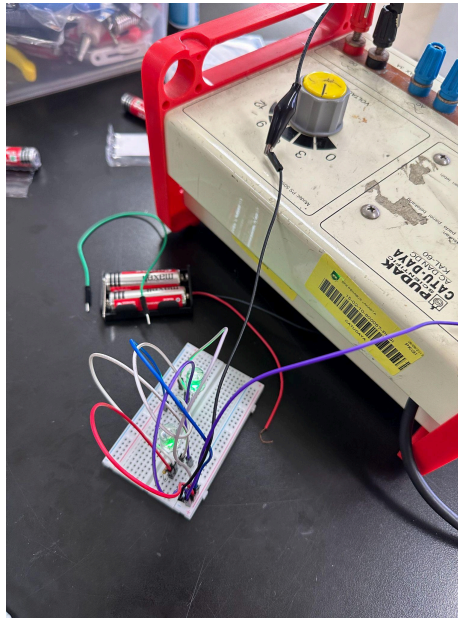
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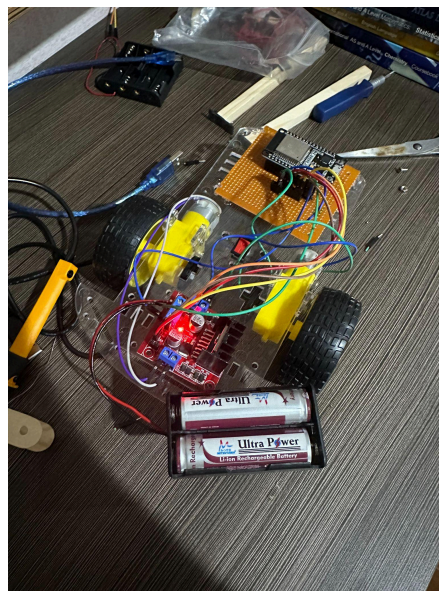
APPENDIX

I.



APPENDIX I. Lab Experiments on One Part of The Solar Charge Controller

II.



APPENDIX II. First Attempt on The Manufacture of The Car Using Batteries As Power Source